

Metadata – Ecological Hive – Series « 2024_06_20_Csv »

Datasets : 2024_12_23_ Environment.csv, 2024_12_23_ EcologicalHive.csv

Apiary location: Mas Lafont, 30170 CROS, France

Hive type: Empty wooden Ecological Hive (or Lunar Hive), (without frames,

frames,

Hive environment: open area

Hives orientation: Bees entrance toward south.

Description:

Ecological Hive (also called Lunar Hive):

- 1 sensor installed under hive roof: central position
- 1 sensor installed on the overframe: south position
- 1 sensor in the upper part of the hive body
- 1 sensor in the lower part of the hive body
- 1 sensor below hive



Environment:

- The luminosity (**6 sensors**) values in each direction (north, south, east, west, top, bottom) is measured on each face of a Dadant close (3m) to the trunk hive.
- Outside air sensor
- Wind speed

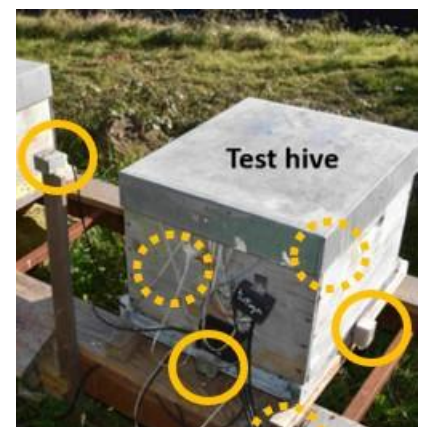


Figure 1: Luminosity sensors

Communications between the acquisition system and the sensors are done via the I2C protocol. If necessary, I2C hubs were inserted to extend number of sensors that can be connected simultaneously.

The sensors provide an internal analog-to-digital conversion. The sensors were individually calibrated to cancel the bias error between them to allow more precise temperature difference measurements (<0.5°C). Absolute temperature errors are probably around 1°C. The intercalibration were achieved by putting all sensors together in an insulated box.

The acquisition period is set to one minute except for the wind speed data which are acquired every 2 s and averaged over 1 min.

Data are stored using a csv format in 2 different files:

- 1) 2024_12_23_Environment.csv
- 2) 2024_12_23_EcologicalHive.csv

Comment lines are prefixed with hash character ‘#’.

List of columns in csv file:

Field name	Description	Unit	Sensor type	Sensor location	
Timestamp	Timestamp of measurement: Year/month/ Day/Hour/ min/sec	(local time)	PCF8523 (Real-time clock)+ DS3231 (Real-time clock module)	Acquisition system	
Epoch [day]	Timestamp of measurement	Day since 1 January 0			
Wind.speed [m/s]	Wind speed	m/s	Sparkfun meteo system	Close to the apiary, 2m high	
Ground.T [deg]	Temperature at the ground surface (≈1cm)	°C	NCT75	On the ground in a shaded area.	
Sky.T [deg]	Equilibrium temperature of a horizontal black steel plane exposed to sunlight and insulated at the rear face	°C	NCT75	Fixed to the weather station	



Figure 2: I2C hubs

the

Luminosity.south.als [lux]	Luminosity south	Lux	VEML7700 (als channel)	South side of the Dadant hive	
Luminosity.south.white [lux]	Luminosity south	Lux	VEML7700 (white channel)	South side of the Dadant hive	
Luminosity.top.als [lux]	Luminosity top	Lux	VEML7700 (als channel)	Top side of the Dadant hive	
Luminosity.top.white [lux]	Luminosity top	Lux	VEML7700 (white channel)	Top side of the Dadant hive	
Luminosity.north.als [lux]	Luminosity north	Lux	VEML7700 (als channel)	North side of the Dadant hive	
luminosity.north.white [lux]	Luminosity north	Lux	VEML7700 (white channel)	North side of the Dadant hive	
Luminosity.west.als [lux]	Luminosity west	Lux	VEML7700 (als channel)	West side of the Dadant hive	
Luminosity.west.white [lux]	Luminosity west	Lux	VEML7700 (white channel)	West side of the Dadant hive	
Luminosity.below.als [lux]	Luminosity below	Lux	VEML7700 (als channel)	Downward facing below the Dadant hive	
Luminosity.below.white [lux]	Luminosity below	Lux	VEML7700 (white channel)	Downward facing below the Dadant hive	
Luminosity.east.als [lux]	Luminosity east	Lux	VEML7700 (als channel)	East side of the Dadant hive	
Luminosity.east.white [lux]	Luminosity east	Lux	VEML7700 (white channel)	East side of the Dadant hive	
Luminosity.horizontal.ALS [lux]	Luminosity horizontal plane	Lux	VEML7700 (als channel)	Upward facing fixed on the weather station	
Luminosity.horizontal.WHITE [lux]	Luminosity horizontal plane	Lux	VEML7700 (white channel)	Upward facing fixed on the weather station	
Luminosity.45.CH0 [lux]	Luminosity on a 45-tilted plane (azimuth=south)	Lux	TSL2561	Fixed on the weather station	
Luminosity.45.CH1 [lux]	Luminosity on a 45-tilted plane	Lux	TSL2561	Fixed on the weather station	

	(azimuth=south)				
Luminosity.45. LUX [lux]	Luminosity on a 45-tilted plane (azimuth=south)	Lux	TSL2561	Fixed on the weather station	
Lunar.roof.T [deg]	Temperature of hive roof	°C	NCT75	Glued under the hive roof, centered	
Lunar.overframe. south.T [deg]	Temperature in hive overframe	°C	NCT75	Overframe south	
Lunar.body.top .T [deg]	Temperature in the upper part of the hive body	°C	NCT75	Inside the hive body, on the top part	
Lunar.body.bot .T [deg]	Temperature in the lower part of the hive body	°C	NCT75	Inside hive body (bottom), in the center	
Lunar.below.T [deg]	Temperature below the hive	°C	NCT75	Centred below the hive floor	